Nandiraju Gireesh

3rd year Undergraduate Student GitHub: github.com/user432

Email: f20170720@hyderabad.bits-pilani.ac.in Mobile:+91-9492901240

Education

Birla Institute of Technology and Science – Pilani, Hyderabad Campus Telangana, India

B.E.(Hons.) in Electronics and Instrumentation *August 2017 – May 2021*

Skills

- **Programming Languages:** Python, C++, Bash
- **Frameworks:** PyTorch, TensorFlow, Keras, fastai, OpenCV, ROS, OpenAI Gym, MuJoCo
- **Tools:** Kubernetes, Docker, GIT, Jupyter, Gazebo, Google Colab

Relevant Courses (Online)

- CS 285 - Deep Reinforcement Learning, Decision Making, and Control – Prof. Sergey Levine (Current)
- Self-Driving Car Engineer Nanodegree – Udacity – (Auditing) (Current)
- Practical Deep Learning for Coders – fast.ai
- Deep Learning from the Foundations – fast.ai

Projects

Vehicle Detection and Tracking Project February 2020

(Object Detection, Computer Vision, Deep Learning, TensorFlow)

- Built a software pipeline which detects vehicles in a video stream and outputs an annotated video stream with a bounding box over vehicles detected.
- Performed a Histogram of Oriented Gradients (HOG) feature extraction on a labeled training set of images and trained them using a Linear Support Vector Machine Classifier (SVM).
- Combined my previous lane-finding algorithm to this project and produced a simultaneous lane-finding and vehicle detected annotated video as output.

Behavior Cloning February 2020

(Imitation Learning, Computer Vision, Deep Learning, TensorFlow)

- Trained a Deep Neural Network using TensorFlow that replicates human steering behavior while driving, thus being able to drive autonomously on a simulator with an accuracy of 98%.
- The model is partially based on NVIDIA's network architecture mentioned in their research paper "End to End Learning for Self-Driving Cars".
- This network takes as input the frame of the frontal camera (say a roof mounted camera) and predicts the steering direction at each instant.

Traffic Sign Recognition Classifier January 2020
(Image Recognition, Computer Vision, Deep Learning, TensorFlow)

- Built and trained a Deep Neural Network which decodes traffic signs from natural images by using German Traffic Sign Dataset.
- The model is based on LeNet Architecture, which is a Convolutional Neural Network designed to recognize visual patterns directly from pixel images with minimal pre-processing.
- Implemented the model using TensorFlow and Scikit-learn's Pipeline framework, which showed an accuracy of 96% on Test Dataset with 500 epochs on CPU.

Rover Search and Sample Return Project February 2020
(Image Processing, Robotics, ROS, Python)

- Developed a model in Python to autonomously map a simulated environment and search for samples of interest.(Based on NASA's sample return challenge)
- Used Color Thresholding and Perspective Transformation methods and trained the model.

Finding Lane Lines on the road for Autonomous Vehicles January 2020
(Image Processing, Computer Vision, Python)

- Built a pipeline that identifies lane lines taking road images from video as input and returns an annotated video as output.
- Applied Canny Edge Detection, Region of Interest Selection and Hough Transform Line Detection techniques and processed the video clips to find lane lines in them.
- Used Python and OpenCV to build the pipeline and the whole model was built in Jupyter Notebook.

Image Classification November 2019
(Computer Vision, Deep Learning, PyTorch, fastai)

- Built a Convolutional Neural Network classifier model that differentiates between 37 different categories of dog & cat breeds.
- Trained my model on Oxford-IIIT PETS Dataset using PyTorch and fastai libraries.
- Used Transfer Learning on my model with ResNet34 & ResNet50 pre-trained models and fine-tuned it to an accuracy of 96%.

Semantic Segmentation November 2019
(Computer Vision, Deep Learning, PyTorch, fastai)

- Built a U-Net architecture model in fastai and PyTorch on CamVid: Motion Based Segmentation and Recognition Dataset.
- Trained the model and obtained pixel-wise segmented images with an accuracy of 93%.

Experience

Summer Internship
RAMCO Cements Pvt. Ltd. May 2019 – July 2019

- During my term as an intern, I have completely studied about the Cement packing process, Bag counting system, SCADA Software, Photo Sensors and Roto-packers that are being used in the plant.
- My aim is to troubleshoot the existing flaws in the current cement packing process and to suggest corrections to increase efficiency of the plant.
- We have introduced Robot Palletizers to increase output efficiency and have reduced the effort put in the whole cement bag packing process.

Scholarships

Prime Minister's Scholarship Scheme (May 2018 - Present)